

DispatchRAG: Grounding Emergency Dispatch Decisions in Real-World Protocols from Traffic Accident Video

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Abstract—Assessing the severity of a traffic accident scenario is important to decide which emergency service to dispatch. Missing an ambulance dispatch on a pedestrian accident is a fatal issue that can lead to death. Recently, Vision-Language Models (VLMs) have been a promising tool for accident reasoning, yet many VLMs are not grounded in real-life accident response protocols, making them not usable in accident severity assessment off-the-shelf. We introduced DispatchRAG, an accident assessor and dispatcher framework grounded in real-life Japanese traffic-accident response protocols, designed to enhance VLMs to generate an appropriate emergency response during an emergency scenario. Utilizing a RAG-based retrieval mechanism to retrieve the most relevant accident protocol and an LLM-powered reasoner to suggest the most proper response. To support evaluation, we introduce Accident Dispatch Dataset, a comprehensive dataset of accident assessment and emergency response according to Japanese accident response protocols adapted from the MM-AU dataset [1]. We validate our framework on the Accident Dispatch Dataset, showing strong performance across various accident scenarios compared to the baseline VLM, pointing toward integration in autonomous vehicles that can automatically report both their own and nearby accidents.

Index Terms—Vision-Language Models, Retrieval-Augmented Generation, Traffic Accident Assessment, Accident Reporter, Autonomous Vehicles

I. INTRODUCTION

In 2023, there were approximately 1.19 million traffic accident deaths [2]. The survivability rate of these victims depends on how quickly emergency services are dispatched. However, the average response time from the accident to a report call is around 4.61 minutes in urban areas and can go up to 7.24 minutes in rural areas [3]. This number is correlated with the fact that rural areas have fewer bystanders to report the accident compared to urban areas. With this gap, deploying an automatic traffic accident responder that can assess the damage severity and suggest the proper emergency response dispatch is needed to shorten the response time, especially in rural areas where no one may be present to call at all. Integrated into an autonomous vehicle, such a responder could report not only the vehicle’s own crash but also accidents it witnesses involving other vehicles, turning every passing vehicle into a potential reporter.

Recent work on traffic accident analysis [4], [5], [6], [7] has begun to leverage Vision-Language Models (VLMs) to

understand and classify traffic anomalies. However, these efforts largely stop at recognizing what happened. The closest to ours is EchoTraffic [8], which moves toward response but only produces general recommendations. It neither assesses the severity of the accident nor decides which emergency units should be dispatched. As a result, no existing work performs the task that an emergency dispatcher does, assessing the urgency of an accident and dispatching the appropriate services accordingly. Moreover, this gap cannot be closed by prompting an off-the-shelf VLM, which lacks knowledge of real emergency protocols. Such models answer only from parametric memory, cannot justify their decisions against any authoritative standard, and are prone to hallucination, which disqualifies them from a safety-critical role. Building a reliable automatic accident responder requires grounding the model’s decisions in real emergency and traffic protocols, which is the approach that we take in this work.

To bridge this gap, we introduce **DispatchRAG**, a retrieval-augmented framework for traffic accident assessment and emergency dispatch decisions. Inspired by the regulation-grounded decision-making of DriveReg [9], DispatchRAG grounds every decision in real emergency protocols rather than in the model’s parametric memory. Given a dashcam clip, the framework (1) converts the scene into a textual description, (2) retrieves the relevant emergency protocols from a curated corpus of Japanese traffic law and EMS guidelines, (3) assesses the severity of the accident against those protocols through a law-grounded urgency guidebook, and (4) recommends actions intended towards three points of view (POVs): the dispatching agency, the on-scene civilian, and the arriving EMS crew.

To evaluate our framework, we further introduce the **Accident Dispatch Dataset**, a new benchmark of **500 dashcam accident clips** built on the videos of the MM-AU dataset [1]. Each clip is annotated manually, blind to any model output and justified rule by rule against the accompanying protocol corpus—with the urgency tier, the emergency units to dispatch (POV 1), the civilian instructions (POV 2), and the crew briefing (POV 3), together with the set of applicable protocol rules.

Our contributions are summarized as follows:

- We propose **DispatchRAG**, the first framework to perform the emergency responder’s task on accident video:

it retrieves the applicable emergency protocols, assesses scene severity according to them, and produces a three-POV dispatch plan grounded in the cited rules.

- We introduce the **Accident Dispatch Dataset**, a benchmark of 500 dashcam accident clips paired with a bilingual corpus of real Japanese traffic-law and EMS protocols, annotated with urgency tiers, dispatch units, and per-POV response plans.
- Experiments show that our decomposed, protocol-grounded design outperforms end-to-end frontier VLMs on the structured dispatch decisions: urgency, units to dispatch, and transport.

II. RELATED WORK

A. Vision-Language Model for Traffic Accident Understanding

Recent work has begun to apply Vision-Language Models to analyze traffic accidents. Benchmark works [1], [5], [7], [10] are providing datasets to evaluate the quality of VLMs in understanding traffic accidents. Recent frameworks proposed classifying accidents type with narrative descriptions [4], fine-grained spatial and temporal grounding of the accidents [6], and adding audio cues to enhance anomaly detection under adversarial conditions, such as nighttime or heavy snow [8]. Collectively, these systems only cover scene understanding and do not produce an actionable emergency-response decision.

B. From Accident Understanding to Emergency Response

A smaller set of research directions moves from accident understanding toward emergency response. GPT-4V evaluates VLM performance on various accident videos and finds that a prompted zero-shot VLM can describe a scene and propose emergency response, such as notifying medical, fire, and police services [11]. Another work combines a detector, a captioner, and a language model to produce incident reports with dispatch suggestions [12]. Other work targets post-collision cause and liability analysis rather than dispatch [13]. In conclusion, these systems are focused more on a dispatch recommendation generated as free-form text without grounding the decision in any emergency-dispatch protocol or legal standard, leaving them unsuitable for a real-life accident dispatching system.

C. Retrieval-Augmented and Protocol-Grounded Dispatch

Two adjacent directions inform our approach. First, retrieval-augmented reasoning has been used to make safety-critical driving decisions trustworthy and auditable. DriveReg retrieves applicable traffic regulations and reasons over them to judge the legality of driving maneuvers [9]. However, it addresses normal-driving compliance rather than accidents or dispatch. Second, automated emergency dispatch triage itself is well studied, but operates only on text. Recent LLM-powered systems infer urgency or priority from dispatch-call transcripts, emergency narratives, and ambulance requests [14], [15], [16]. These methods prioritize a request already described by a human caller, rather than perceiving the accident itself. In contrast, our work makes the dispatch decision directly from visual evidence of the accident scene and grounds each decision in

retrieved rules of real emergency and traffic protocols with an explicitly cited rule, producing an auditable dispatch decision that prior work does not provide.

III. PROPOSED METHOD

A. Overview

Given a dashcam clip of a traffic accident, our goal is to produce an emergency-dispatch plan in which every decision is grounded in an authoritative Japanese law or medical protocol. End-to-end VLMs can describe a crash, but they cannot state which rule justified a decision. As shown in Fig. 1, our framework therefore decomposes the task into three stages: a *Perception* stage that turns the clip into a free-text scene description, a *Rule-Retrieval Agent* that fetches the applicable rules from a curated regulatory corpus, and a *Reasoning Agent* that composes a three-perspective (3-POV) dispatch plan grounded in those rules. The 3-POV plan targets the responding agency, the on-scene civilian, and the arriving EMS crew. Each directive carries the cited rule that justifies it. Video is consumed only at perception; all downstream decisions operate on text and cited rule identifiers, so the plan can be checked against its sources.

B. Japanese Legal and Protocol Corpus

The corpus consists of 19 Japanese rule passages curated from authoritative Japanese sources, which are the Road Traffic Act and Civil Code, the Fire and Disaster Management Agency’s emergency service guidance (including the four-tier urgency classification), regional medical-control trauma protocols, and the Japan Resuscitation Council’s basic life support guidelines [17], [18], [19], [20], [21], [22]. Combined, they cover the driver’s legal duties, urgency assessment, unit dispatch, and prehospital care.

C. Perception

The perception VLM receives the 16 sampled accident frames and writes a natural-language paragraph describing the accident details. It includes collision type, vehicles involved, visible hazards (deformation, persons down, debris, vehicle rollover, fluid leak, fire), each vehicle’s damage severity, and the scene environment. The prompt only includes an observable scene description without including inferred information, such as clinical state. This is meant to keep the reasoning process from being biased by these keywords.

D. Rule-Retrieval Agent

Since the VLM only describes the scene objectively, it does not contain any actionable vocabulary that is stored in the rule corpus, which means a direct rule retrieval would miss important documents. Utilizing the HyDE paradigm [23], an LLM rewrites the observation into a hypothetical professional description of what this scene might require, such as the casualty’s severity assessment, transport urgency, driver’s legal duties, and scene-safety concerns. This information is inferred without citing any article number or document titles. This enables the query to retrieve action-oriented rules.

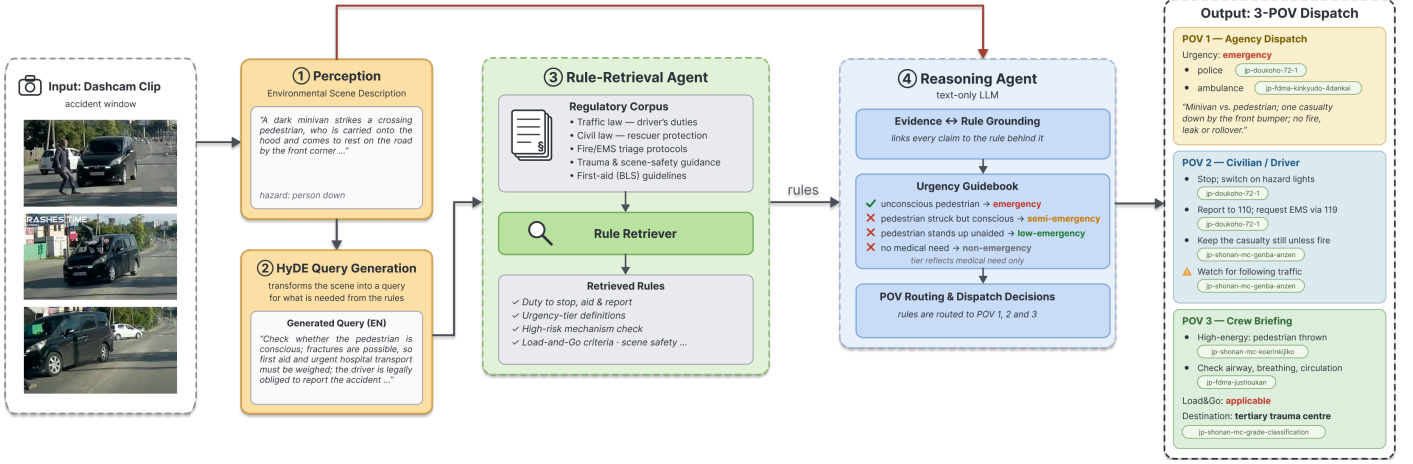


Fig. 1. Overview of the proposed framework. A VLM converts the dashcam clip into a prose scene description, which is rewritten into an actionable retrieval query. The Rule-Retrieval Agent matches the query against a bilingual corpus of Japanese statutes and EMS protocols. The Reasoning Agent grounds each claim in observed evidence and a rule identifier, assigns the urgency tier via a law-grounded guidebook, and routes content to three audiences, producing a 3-POV dispatch plan.

The query and the corpus passages are encoded by a multilingual sentence-embedding model [24], and the agent retrieves the top- k passages by cosine similarity to form the rule set passed to the reasoner.

E. Reasoning Agent

The Reasoning Agent is a text-only LLM that receives the scene description and the retrieved rules and writes the dispatch plan. Three mechanisms make the output auditable and reproducible.

1) *Evidence–rule grounding*: The agent may cite only law identifiers from the retrieved rules, and every claim must be paired with the observed evidence that supports it; conditional phrasing is reserved for genuinely unobservable facts (e.g., internal injury).

2) *Law-grounded urgency guidebook*: The system maps observable cues onto the four-tier severity scale of the 119 emergency-call protocol, each tier carrying its dispatch implication. The *emergency* tier (Red) covers the most time-critical conditions: suspected cardiopulmonary arrest, an unconscious struck pedestrian, and high-energy mechanisms such as vehicle fire or entrapment that demand rapid arrival and transport. The *semi-emergency* tier (Yellow) covers conditions still needing medical judgment with less time pressure, requiring a hospital visit within about two hours. The *low-emergency* tier (Green) covers conditions outside the first two that still require examination, and the *non-emergency* tier (White) covers conditions needing no medical attention. The tier reflects medical need only, and forms an ordered scale (non-emergency < low-emergency < semi-emergency < emergency); over- and under-triage denote predicting a tier above or below the ground truth.

3) *POV routing*: The produced content is routed to three audiences with different needs:

- **POV 1 — Agency dispatch** (119/110 dispatcher): machine-actionable fields, such as the urgency tier, the

emergency units to dispatch (police, ambulance, fire-rescue, doctor-car) each with a cited reason and an incident summary.

- **POV 2 — On-scene civilian** (driver, bystanders): short imperative do/warn/don’t directives executable without medical training: driver duties during an accident, first aid to give or withhold, and scene-safety warnings.
- **POV 3 — Arriving EMS crew**: injury-mechanism summary, arrival checklist (consciousness assessment, spinal precaution, injury assessment), the Load-and-Go decision, and the medical service destination.

F. Dispatch Dataset and Annotation

We construct the **Accident Dispatch Dataset** by annotating 500 dashcam accident clips from MM-AU [1] with the full 3-POV dispatch plan. The clips span a range of accident types, including vehicle–vehicle, vehicle–motorcycle, vehicle–bicycle, vehicle–pedestrian, vehicle–object, and single-vehicle accidents. All clips are annotated manually, blind to any model output, with every field justified by the cited corpus rules. Each annotation collects four field groups:

- **Observation tags** — the objective visible facts: collision type, involved vehicles, environment, scene hazards, damage severity, and the post-impact state of a struck vulnerable road user (unconscious/not moving, moving but unable to stand, stands up unaided, not visible).
- **Urgency tier** — assigned with the same law-grounded guidebook used by the reasoner (Sec. III-E), keeping the label grounded in the protocol rather than guessed.
- **Structured 3-POV plan** — the ground-truth dispatch in the same schema that the system produced, every item is cited.
- **Applicable rules** — the rule identifiers that ground the clip’s dispatch decisions.

Table I positions our dataset against prior traffic-video benchmarks. These existing datasets are focused on explaining

what happened. EchoTraffic provides audio-visual QA pairs for anomaly understanding [8], MM-AU annotates object boxes, accident time windows, and causes for abductive reasoning [1], SUTD-TrafficQA provides multiple-choice questions across six event-reasoning tasks [5], and VRU-Accident and Crash-Sight pair accident clips with QA and dense-caption annotations [7], [10]. None of them labels the severity assessment and emergency dispatch decision. While several of these datasets are larger, ours is the first to annotate each clip with an actionable dispatch decision, a four-tier urgency label, the emergency units to send, and per-POV instructions. Ours is also the first to ground every label in cited real-world legal and medical protocols, making the annotations grounded rather than subjective.

TABLE I
THE ACCIDENT DISPATCH DATASET VERSUS PRIOR TRAFFIC-VIDEO BENCHMARKS. URG. = URGENCY-TIER LABEL; DISP. = EMERGENCY-UNIT DISPATCH LABEL; LAW = DECISIONS GROUNDED IN CITED LEGAL/MEDICAL PROTOCOLS.

Dataset	Annotation	#Clips	Urg.	Disp.	Law
EchoTraffic [8]	audio-visual QA	29,865	×	×	×
MM-AU [1]	boxes, time, cause	11,727	×	×	×
SUTD-TrafficQA [5]	event-reason. QA	10,080	×	×	×
VRU-Accident [10]	VQA + caption	1,000	×	×	×
CrashSight [7]	VQA + caption	250	×	×	×
Ours	3-POV dispatch	500	✓	✓	✓

IV. EXPERIMENTS

A. Experimental Setup

1) *Dataset*: The Accident Dispatch Dataset consists of 500 annotated clips prepared under the protocol of Sec. III-F; we report results on the whole dataset. Scoring covers all ground-truth clips.

2) *Metrics*: For the structured decisions: *Unit F1* on the dispatched unit set; *Urgency EM*, exact match on the four-tier label, and *Urgency Ord.*, an ordinal agreement score ($1 - |g - p|/3$ over the ranked tiers) that credits near-miss tiers; and *Load-and-Go Accuracy* on the POV 3 transport decision. Because urgency is ordinal, we also report the directional error rates *over-triage* (predicting a tier above the ground truth) and *under-triage* (predicting below); a well-calibrated system minimizes both rather than trading one for the other.

We separately analyze the binary ambulance decision (dispatch or not) with accuracy, precision, recall, and F1, since recall alone can be inflated by indiscriminately dispatching to every clip. Content quality uses an LLM judge (per-POV coverage of ground-truth directives). To avoid single-judge and self-preference bias, every judge’s score is the mean of a three-judge panel drawn from three different model families (Gemma-2-27B, Llama-3.3-70B, and Pixtral-12B), none of which is a system under test.

3) *Baseline models*: We compare four methods. **Framework (ours)** is the decomposed pipeline of Sec. III. The baselines are end-to-end *VLMs with Retrieved Rules*: each receives the sampled frames and the same retrieved rules and

emits the 3-POV plan in one call, instantiated with Qwen3-VL-32B[25], GPT-5.4-mini, and Gemini-2.5-Flash[26]. Since all methods see the identical rule context, the comparison isolates the perception/architecture split.

4) *Implementation details*: Perception uses GPT-5.4-mini over 16 uniformly sampled frames; the retriever uses BGE-M3[24] dense embeddings with a fixed top- k budget; the reasoner is Qwen3-32B [27] with the urgency guidebook embedded in its system prompt.

B. Main Results

1) *Structured dispatch results*: On the structured decisions (left of Table II), the framework leads every dispatcher-facing call: unit selection, urgency, and transport, compared to the end-to-end baselines. We explain each result below.

2) *Urgency tier*: Urgency is the hardest field, since it decides every other decision. On the exact-match accuracy (EM), the framework reaches 0.47 against 0.23–0.35 for the baselines, but an exact-match alone is harsh. Therefore, we add a comparison by providing an ordinal scale value, so an off-by-one call is a near miss rather than an outright error. The ordinal score (Ord. 0.77 vs. 0.63–0.70), which gives a full score for an exact tier and degrades linearly with tier distance, showing that even when the framework misses the exact tier, it lands closer on the scale.

It is also the best calibrated: the framework over-triages on only 28% of clips versus 43–60% for the baselines (O-Tri column). The baselines hold under-triage low only by escalating almost everything, a non-discriminating strategy whose cost surfaces in the ambulance analysis below (Table III), where it yields up to three times more false positives. The framework instead balances both error directions.

3) *Ambulance dispatch: calibration over recall*: The most important call is whether to dispatch an ambulance at all. Table III scores this binary over all 500 clips (320 require an ambulance, 180 do not). The framework’s higher under-triage might seem unsafe compared to the baselines’ near-perfect recall, but the confusion counts show why that recall is misleading: the baselines reach up to 0.99 only by dispatching in almost every clip. Gemini-2.5 predicts no ambulance on just 4 of 180 clips and produces 176 false positives, approximately three times the framework’s 58. Recall is trivially maximized by always escalating. Thus, the important objective should be calibration, in which the framework trades a little recall (0.81) for far higher precision (0.82), giving the best accuracy (0.76) and F1 (0.81) score. This implies the framework is the safest and best-balanced method compared to others that send an ambulance to every situation.

4) *Emergency unit dispatch and transport*: On the set of dispatched units, the framework attains the highest Emergency Unit F1 (0.86 vs. 0.73–0.82). The transport decision is evaluated by Load-and-Go, the POV3 prehospital call to transport the patient to the hospital immediately, performing only life-saving interventions on the scene, rather than treating in place, reserved for high-energy or time-critical injuries. The

TABLE II
STRUCTURED DISPATCH DECISIONS AND LLM-JUDGE CONTENT QUALITY ($n = 500$). EM = EXACT MATCH; ORD. = ORDINAL AGREEMENT, MEAN $1 - |g - p|/3$ OVER THE RANKED TIERS; O-TRI/U-TRI = OVER-/UNDER-TRIAGE RATE; COVERAGE = RECALL OF GROUND-TRUTH DIRECTIVES PER POV, AVERAGED OVER A THREE-JUDGE PANEL (GEMMA-2-27B, LLAMA-3.3-70B, PIXTRAL-12B)

Method	Structured dispatch decisions						Judge coverage		
	Unit	Urgency				L&G			
	F1 ↑	EM ↑	Ord. ↑	O-Tri ↓	U-Tri ↓	Acc ↑	POV1 ↑	POV2 ↑	POV3 ↑
Qwen3-VL-32B + Rules	0.82	0.35	0.68	0.51	0.13	0.38	0.43	0.47	0.50
GPT-5.4-mini + Rules	0.73	0.35	0.70	0.43	0.22	0.39	0.56	0.67	0.66
Gemini-2.5 + Rules	0.79	0.23	0.63	0.60	0.17	0.32	0.43	0.52	0.51
Framework (ours)	0.86	0.47	0.77	0.28	0.25	0.55	0.55	0.56	0.58

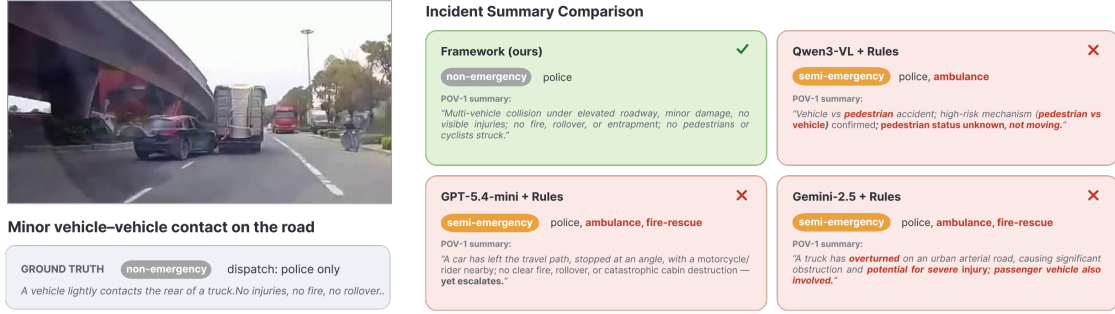


Fig. 2. On a minor vehicle-vehicle contact (GT: *non-emergency*, police only), the rule-augmented VLMs over-triage. Qwen3-VL-32B and Gemini-2.5 are hallucinating a severe element absent from the clip, while GPT-5.4-mini is escalating despite an accurate and benign description. It predicts unnecessary units. The framework holds the correct tier and dispatch.

TABLE III
AMBULANCE-DISPATCH DECISION (BINARY, $n = 500$; 320 REQUIRE AN AMBULANCE, 180 DO NOT). TP/TN = CORRECT DISPATCH/WITHHOLD; FP = UNNECESSARY DISPATCH (OVER-TRIAGE); FN = MISSED DISPATCH (UNDER-TRIAGE).

Method	Acc $\uparrow$	Rec. $\uparrow$	Prec. $\uparrow$	F1 $\uparrow$	TP/TN/FP/FN
Qwen3-VL-32B + Rules	0.72	0.92	0.72	0.81	295/65/115/25
GPT-5.4-mini + Rules	0.69	0.97	0.68	0.80	311/33/147/9
Gemini-2.5 + Rules	0.64	0.99	0.64	0.78	316/4/176/4
Framework (ours)	0.76	0.81	0.82	0.81	258/122/58/62

framework makes this call best as well (L&G Acc 0.55 vs. 0.32–0.39).

5) *Content quality*: On the prose quality (right of Table II), the result separates cleanly. GPT-5.4-mini with retrieved rules writes the richest directives, leading judge coverage across all three POVs, while the framework’s text is terser and more structured. The two are complementary: the framework wins the *structured* dispatch, while a frontier VLM writes the richer *narrative* crew and civilian content.

TABLE IV
RECALL OF THE GROUND-TRUTH CITED RULES WITH PERCEPTION AND REASONER HELD FIXED. ONLY THE QUERY/SCORER VARIED.

Retrieval method	Recall $\uparrow$
BM25 [28]	0.71
Direct Observation	0.73
HyDE (ours)	0.76

6) *Retrieval ablation*: Table IV isolates the query formulation, holding perception and the reasoner fixed and varying only how the corpus is queried. The HyDE recalls the ground-truth cited rules best (0.76), ahead of a query built directly from the raw observation (0.73) and lexical BM25 (0.71). This suggests the HyDE paradigm transforms the query into an action-oriented register of the rule text, closing the vocabulary gap of Sec. III-D.

7) *Limitations*: Two limitations remain. First, the framework trades recall for precision on the ambulance call: it is the best-calibrated method overall, but a small fraction of true ambulance cases are under-triaged, and raising recall without re-inflating false positives is an open problem. Second, perception operates on a single forward dashcam view, so occluded or out-of-frame casualties can be missed; a multi-view or sensor-fused input would relax this constraint.

C. Qualitative Comparison

Fig. 2 illustrates that the baselines over-triage a non-emergency case of a minor car-to-truck contact. They fail in two ways: Qwen3-VL-32B and Gemini-2.5 hallucinate a severe element absent from the scene, a phantom pedestrian and an overturned truck, while GPT-5.4-mini describes the scene accurately, correctly noting no fire or rollover, yet still escalates and over-dispatches. In both modes, the framework’s rule-grounded tiering keeps the call grounded in what is actually visible.

V. CONCLUSION

We presented DispatchRAG, a retrieval-augmented framework that turns a dashcam accident clip into a law-grounded, three-perspective emergency dispatch plan, together with the Accident Dispatch Dataset, the first benchmark to annotate accident video with an actionable response, urgency tier, dispatched units, and per-POV instructions grounded in cited Japanese legal and medical protocols.

Our experiments support two conclusions. First, decomposition delivers the most reliable critical decisions. Separating perception from rule-grounded reasoning yields the most accurate urgency, unit, and transport decisions and the best-calibrated triage, avoiding the over-triage that escalates the end-to-end baselines, including newer frontier VLMs running end-to-end. Second, the two methods are complementary. Our framework produces an auditable, well-calibrated structured plan with rule-level citations, while a frontier VLM writes richer free-text sentences.

Several directions remain for future work. First, we plan to expand the dataset with more annotated clips to cover a wider range of accident scenarios. Second, we aim to extend the corpus beyond Japanese protocols to the emergency and traffic regulations of other jurisdictions, broadening the framework’s applicability. Third, we intend to expand the framework in a multi-vehicle setting, enabling collaborative perception that can assess damages from multiple views, addressing the occlusion limitation of a single camera viewpoint. We will release the dataset and corpus on GitHub later to support further work on grounded, accountable emergency-response systems.

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